ZEPRALAN 1000 MG/G SOLUBLE POWDER

FOR ANIMAL USE ONLY
PRODUCT NAME:
ZEPRALAN 1000 MG/G SOLUBLE POWDER
NAME AND STRENGTH OF ACTIVE INGREDIENT:
Each g contains Apramycin Sufate 1000 mg
PRODUCT DESCRIPTION
A light brown to yellowish powder or granular powder. Each g contains Apramycin Sufate 1000 mg
PHARMACODYNAMIC PROPERTIES:
Apramycin is a broad-spectrum bactericidal aminoglycoside antibiotic whose action results from the binding to the 30S subunit of the ribosome, preventing protein synthesis and disrupting the membrane permeability of bacteria. Apramycin is effective against Gram-negative bacteria (*Salmonella* and *Escherichia coli*). Resistance mechanisms: Different aminoglycoside 3-N acetyltransferase enzymes (AAC-3) have been related with resistance to apramycin. These enzymes confer different cross-resistance against other aminoglycosides. Some strains of *Salmonella* Typhimurium DT104 in addition to resistance against betalactams, streptomycin, tetracyclines and sulphonamides carry a conjugative resistance plasmid against apramycin. Apramycin resistance can be influenced by co-selection (resistance to apramycin has been described to be located in the same mobile genetic element that other resistant determinants in *Enterobacteriaceae*) and cross resistance (*e. g.* with gentamicin). Resistance developed by chromosomal resistance is minimal for most of the aminoglycosides.
PHARMACOKINETIC PROPERTIES:
The oral administration of apramycin is intended for antimicrobial activity within the gut; apramycin is poorly absorbed, but absorption may be increased in young animals and in animals with disrupted intestinal barrier. Absorption: Absorption may be high in new-born animals but rapidly decreases in the first weeks of life. Distribution, biotransformation and excretion: Apramycin is mainly excreted through faeces, under active form, and only a small quantity is excreted in the urine. Very little metabolism of apramycin takes place in the pig.
INDICATIONS:
For the use in the control and treatment of colisepticaemia (colibacillosis) caused by susceptible strains of *E. coli* in broilers.
TARGET SPECIES
Chickens (broiler)
ROUTE OF ADMINISTRATION
To be administered via the drinking water.
DOSAGE
Broiler: 100 g – 165 g product per 200 L drinking water (equivalent to 250 to 500 mg Apramycin activity / L of drinking water) for 5 consecutive days. This is equivalent to an average of 40 – 80 mg Apramycin activity per kg body weight daily.
DIRECTION OF USE
Drinking systems should be clean and free of rust to avoid reduction in activity. Add the appropriate quantity of product to 5 times its own volume of water. Stir well and allow solution to stand for a few minutes then stir again. When completely dissolved, add to the full volume of drinking water and stir well. Prepare fresh medicated drinking water every 24 hours to ensure activity.
CONTRAINDICATIONS
Do not use in case of hypersensitivity to apramycin.
SPECIAL WARNINGS FOR EACH TARGET SPECIES
None.
SPECIAL PRECAUTION FOR USE
Special precautions for use in animals:
Use of the veterinary medicinal product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria. Where a diagnosis of *Salmonella* Dublin is made on the farm, then control measures including on-going monitoring of disease status, vaccination, biosecurity and movement controls should be considered. National control programmes should be followed where available. Use of the veterinary medicinal product deviating from the instructions given in the label may increase the prevalence of bacteria resistant to the apramycin and may decrease the effectiveness of treatment with aminoglycosides due to the potential for cross-resistance. Official, national and regional antimicrobial policies should be taken into account when the veterinary medicinal product is used.
Special precautions to be taken by the person administering the veterinary medicinal product to animal:
People with known hypersensitivity to apramycin or any other aminoglycoside should avoid contact with the product. This product may cause irritation or sensitisation after skin or eye contact or inhalation. Avoid contact with the eyes, skin and mucous membranes and inhalation of dust while preparing the medicated water/milk replacer. Use personal protective equipment consisting of gloves, mask, goggles and protective clothing while handling the product. Wash hands after use. In case of eye contact, rinse the affected area with plenty of water. In case of skin contact, wash thoroughly with soap and water. If irritation persists, seek medical advice. In the case of accidental ingestion, seek medical advice immediately and show the package leaflet or the label to the physician. In case of onset of symptoms after exposure such as skin rash, seek medical advice immediately and show the package leaflet or the label to the physician. Swelling of the face, lips and eyes or difficult breathing are more serious symptoms and require urgent medical assistance.
INTERACTIONS WITH OTHER MEDICAMENTS
Aminoglycosides may have a negative influence on the kidney function. The administration of aminoglycosides to animals suffering from renal impairment or in combination with substances that also affect renal function may therefore present a risk of intoxication. Aminoglycosides may cause neuromuscular blockade. It is therefore recommended to take such an effect into account when anaesthetising treated animals.
USE DURING PREGNANCY, LACTATION AND LAY
Chickens: Do not use in laying hens.
ADVERSE EFFECTS
None known
OVERDOSE AND TREATMENT
Chicken: There was no mortality when chickens were given a single oral dose of 1,000 mg/kg of bodyweight. Chickens were given up to 5 times the recommended level for 15 days with no untoward reaction. Possible intoxications can be recognised by the following symptoms: soft faeces, diarrhoea, vomiting (weight loss, anorexia, and similar), renal impairment and effects on the central nervous system (reduced activity, loss of reflexes, convulsions, etc.). Do not exceed the recommended dose.
WITHDRAWAL PERIODS
Chickens (Meat): 14 days. Eggs: Do not use in chickens which are producing or may in the future produce eggs or egg products which may be used or processed for human consumption.
SHELF-LIFE
Shelf life as packaged for sale: 2 year. Shelf life after first opening : 24 hours. Shelf life after reconstitution : 24 hours
ALLOWABLE MAXIMUM RESIDUAL LIMIT (MRL)

Food	MRLs (ppm)	MRLs (µg/kg)	Reference
Chicken			
Muscle	0.5	500	The Japan Food Chemical Research Foundation
Fat	0.5	500	The Japan Food Chemical Research Foundation
Liver	0.5	500	The Japan Food Chemical Research Foundation
Kidney	2	2000	The Japan Food Chemical Research Foundation
Edible Ofal	2	2000	The Japan Food Chemical Research Foundation

STORAGE CONDITIONS
Store in a cool place below 30ºC. Keep out of reach of children. Jauhi ubat daripada kanak-kanak.
DISPOSAL OF CONTAINERS
Any unused veterinary medicinal product or waste material derived from such veterinary medicinal products should be disposed of in accordance with local requirements.
PACKAGING AVAILABLE
100 g, 200 g, 500 g, or 1000 g per aluminum sachet.
MANUFACTURER AND PRODUCT REGISTRATION HOLDER
Life Biopharma Sdn. Bhd. (company no. 546163-D)
No. 250-251, Jalan Industri Galla 13, Kawasan Perindustrian Galla, 70200 Seremban, Negeri Sembilan, Malaysia.
DATE OF REVISION OF PACKAGE INSERT
12 October 2021